MATH 250 — Eigenvalues & Eigenvectors

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Learning Objectives

- Find eigenvalues by solving the characteristic equation det(A – λI) = 0
- Compute eigenvectors by solving (A – λI)v = 0
- Recognize repeated and complex eigenvalues
- Connect eigenvalues to trace, determinant, and diagonalization

For eigenvalues: factor the characteristic polynomial completely. For eigenvectors: row reduce (A – λI) and express solutions in terms of free variables.

1. Find the eigenvalues of A by solving det(A – λI) = 0.

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Answer: _____

2. Find the eigenvectors corresponding to each eigenvalue of A.

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

Answer: _____

3. Find the eigenvalues of A. Are they real or complex?

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Answer: _____

4. Find the eigenvalues and eigenvectors of A.

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Answer: _____

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5. Find all eigenvalues of A (hint: A is upper triangular).

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

Answer: _____

6. Given $\lambda = 1$ is an eigenvalue of A, find the corresponding eigenvector.

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

Answer: _____

7. Use trace and determinant to find eigenvalues of A without expanding fully.

$$A = \begin{bmatrix} 5 & 2 \\ 1 & 4 \end{bmatrix}$$

Answer: _____

8. Determine if A is diagonalizable. If yes, find matrices P and D such that $A = PDP^{-1}$.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Answer: _____

9. Given A has eigenvalues $\lambda_1 = 2$, $\lambda_2 = -1$, find the eigenvalues of A^3.

$$A\vec{v} = \lambda\vec{v} \Rightarrow A^3\vec{v} = \lambda^3\vec{v}$$

Answer: _____

10. Find the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$ and the eigenvalues.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix}$$

Answer: _____

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Problems 3–4: complex and repeated eigenvalues — connect to geometry. Problem 7: shows the trace/det shortcut. Problem 10: companion matrix characteristic polynomial (advanced).

Solutions

1. Find the eigenvalues of A by solving det(A – λI) = 0.

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

→ *det(A–λI) = det([[4–λ,1],[2,3–λ]]) = (4–λ)(3–λ)–2 = 0.*
→ *λ²–7λ+10 = 0 → (λ–5)(λ–2) = 0.*
→ *Eigenvalues: λ = 5 and λ = 2.*

Answer: $\lambda_1 = 5, \quad \lambda_2 = 2$

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2. Find the eigenvectors corresponding to each eigenvalue of A.

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

→ *Eigenvalues (upper triangular): λ=3, λ=2 (diagonal entries).*
→ *For λ=3: (A–3I)v=0 → [[0,1],[0,–1]]v=0 → v = [1,0]ᵀ.*
→ *For λ=2: (A–2I)v=0 → [[1,1],[0,0]]v=0 → v = [–1,1]ᵀ.*

Answer: $\lambda = 3 : \vec{v} = (1, 0)^T, \quad \lambda = 2 : \vec{v} = (-1, 1)^T$

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3. Find the eigenvalues of A. Are they real or complex?

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

→ *det(A–λI) = (0–λ)(0–λ)–(–1)(1) = λ²+1 = 0.*
→ *λ² = –1 → λ = ±i.*
→ *Complex eigenvalues indicate A is a rotation matrix (90°).*

Answer: $\lambda = \pm i$ (complex, no real eigenvalues)

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4. Find the eigenvalues and eigenvectors of A.

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

→ *det(A–2I) = det([[0,1],[0,0]]) = 0. λ=2 (multiplicity 2).*
→ *(A–2I)v = 0: [[0,1],[0,0]]v=0 → v_2=0, v_1 free.*
→ *Only one independent eigenvector: v = [1,0]ᵀ. (defective matrix)*

Answer: $\lambda = 2$ (repeated), $\vec{v} = (1, 0)^T$

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5. Find all eigenvalues of A (hint: A is upper triangular).

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

→ *For a triangular matrix, eigenvalues = diagonal entries.*
→ *$\lambda_1 = 2$, $\lambda_2 = -1$, $\lambda_3 = 3$.*

Answer: $\lambda_1 = 2,\ \lambda_2 = -1,\ \lambda_3 = 3$

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6. Given $\lambda = 1$ is an eigenvalue of A, find the corresponding eigenvector.

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

→ *$(A-I)v = 0$: $A-I = [[1,-1,0],[0,0,0],[1,-1,0]]$.*
→ *Row reduce: $[[1,-1,0],[0,0,0],[0,0,0]]$.*
→ *x_2 free. $x_1 = x_2$. x_3 free from row 3 (already zero).*
→ *Eigenvectors: $v = t[1,1,0]^T$.*

Answer: $\vec{v} = t(1, 1, 0)^T$

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7. Use trace and determinant to find eigenvalues of A without expanding fully.

$$A = \begin{bmatrix} 5 & 2 \\ 1 & 4 \end{bmatrix}$$

→ *$tr(A) = \lambda_1+\lambda_2 = 5+4 = 9$.*
→ *$det(A) = \lambda_1\lambda_2 = 20-2 = 18$.*
→ *Characteristic eq: $\lambda^2-9\lambda+18=0 \rightarrow (\lambda-6)(\lambda-3)=0$.*
→ *$\lambda_1 = 6$, $\lambda_2 = 3$.*

Answer: $\lambda_1 = 6,\ \lambda_2 = 3$

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8. Determine if A is diagonalizable. If yes, find matrices P and D such that $A = PDP^{-1}$.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

→ *A is already diagonal with eigenvalues $\lambda_1=1$, $\lambda_2=2$.*
→ *Standard basis vectors are eigenvectors: $e_1=[1,0]^T$, $e_2=[0,1]^T$.*
→ *$P = I$ (identity), $D = A$. So $A = I \cdot A \cdot I^{-1} = A$. ✓*

Answer: A is diagonal: $P = I,\ D = A$

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9. Given A has eigenvalues $\lambda_1 = 2$, $\lambda_2 = -1$, find the eigenvalues of A^3.

$$A\vec{v} = \lambda\vec{v} \Rightarrow A^3\vec{v} = \lambda^3\vec{v}$$

→ *If $Av = \lambda v$, then $A^2v = \lambda^2 v$ and $A^3v = \lambda^3 v$.*
→ *Eigenvalues of A^3 are λ^3 for each eigenvalue λ of A.*
→ *$2^3 = 8$, $(-1)^3 = -1$.*

Answer: $\lambda_1^3 = 8,\quad \lambda_2^3 = -1$

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10. Find the characteristic polynomial p(λ) = det(A – λI) and the eigenvalues.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix}$$

→ *det(A–λI): expand along first column.*

→ *This is a companion matrix for p(λ) = λ³–4λ²+5λ–2.*

→ *Try λ=1: 1–4+5–2=0. ✓ Factor: (λ–1)(λ²–3λ+2) = (λ–1)²(λ–2).*

→ *Eigenvalues: λ=1 (multiplicity 2), λ=2.*

Answer: $p(\lambda) = -\lambda^3 + 4\lambda^2 - 5\lambda + 2$

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